

A NOVEL HPLC METHOD FOR THE DETERMINATION OF LEVOFLOXACIN IN BLOOD AND ITS BIOEQUIVALENCE STUDIES

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ABSTRACT:

Bioequivalence study was performed for Levofloxacin Tablets 250mg. For this purpose one reference formulation manufactured in Pakistan, by a Local Pharmaceutical Company which is also a brand leader, was compared with test formulation manufactured by another Pharmaceutical Industry.

The study was formatted in accordance to the draft guidance provided by FDA (CDER) for bioavailability and bioequivalence studies. The test formulation was administered orally, in fasting state, to 7 healthy volunteers and the blood samples were drawn at appropriate intervals upto 30hrs, Reference product (Cravit) was administered to the same group of volunteer after a wash out period of one week.

Samples were analysed on HPLC for the determination of drug concentrations. The Pharmacokinetic parameter such as C_{max}, T_{max} and AUC were calculated. The data generated was analysed statistically by ANOVA and paired t test.

Results of Pharmacokinetic parameter indicated that both products Test (A) and Reference (B) are Bioequivalent. Statistical analysis supports this conclusion.

INTRODUCTION

Levofloxacin is the optical S-(-)-isomer of ofloxacin (Davis R and Bryson H.M., 1994); It is twofold more active than Ofloxacin and tends to be about fourfold more potent than Ciprofloxacin (Gootz T.D. and McGuirk P.R. (1994). Levofloxacin penetrates rapidly into body tissues. Following oral administration of 50-200 mg Levofloxacin, mean maximum plasma concentrations ranging from 0.57 to 2.04 mg/l were achieved within 0.8 to 2.4 hours; these were linearly related to the dose (Davis R and Bryson H.M., 1994).

All fluoroquinolones are well absorbed; reach peak serum concentrations within 1-2 hr (Sparfloxacin in 4-5 hr), and have high distribution volumes (1.5-4.5 liters/kg) typical of the class. Protein binding is generally <50%, although somewhat higher (70%) for

Trovafloxacin. After oral administration the fluoroquinolones produce levels well in excess of those required for clinical efficacy in genitourinary, gynecological, skeletal, and other important tissues and fluids (Gerding D.N. *et al.*, 1989),

Serum concentration of Levofloxacin in healthy adults rose rapidly after oral administration. The time to maximum serum concentration ranged from 0.92 to 1.48 hours after administration, and the maximum serum concentration and area under the serum concentration curve were dose dependent. Levofloxacin does not accumulate in the serum after multiple-dose administration (Nakashima M., 1992).

Levofloxacin has many important advantages based upon its pharmacological profile. These include the fact that it can be given over as an oral compound, using a once

daily administration schedule. It is rapidly absorbed via the gastrointestinal tract and penetrates very rapidly into tissues where it achieves a high concentration, even higher than in the plasma (Davis R and Bryson H.M., 1994).

MATERIALS AND METHOD

Subjects

Volunteers comprised of healthy male subjects aged between 27 and 38 years. All volunteers were informed about the purpose and method of study. Instructions for avoiding drugs and maintaining the fasting state on the morning of the test date, were given in details. Written consent was obtained from each volunteer.

All volunteers were examined by a registered medical practitioner before being included in the study for the purpose of determining their health status.

Study Design

Collection of Samples and Storage:

Bioequivalent study was performed for two different formulation of Levofloxacin on 7 healthy subjects. Serial blood samples were

drawn at 0, 0.5, 1, 1.5, 2, 3, 4, 8, 12, 24 and 30 hrs after administration of drug. Blood was drawn using aseptic procedure in 10ml sterile syringes and transferred immediately to heparinized centrifuge tubes. Plasma was separated from each sample by centrifugation at Hereus centrifuge at 3000 rpm for 5 minutes. Plasma samples were properly marked and stored at -20 C. till analysed.

Assay Method and validation:

A high performance liquid chromatographic method was used for the analysis of Levofloxacin.

A. Equipment Details:

1. HPLC: Shimadzu, LC 10A (Pump)
SPD 10A (Detector)
2. Column: Shimpack CLC ODS (C18)
6.0mmID x 15cm
3. Communication Bus Module (CBM 101, Shimadzu)
4. Pentium II 333 MHz
5. LC 10A (Software for data handling)

B. Mobile Phase Preparation:

Acetonitrile: Methanol: 90mM KI111³0₄ :
Trichloroacetic acid, was mixed at a ratio of
15:5:80:0.3

C. Column Temperature: 35°C

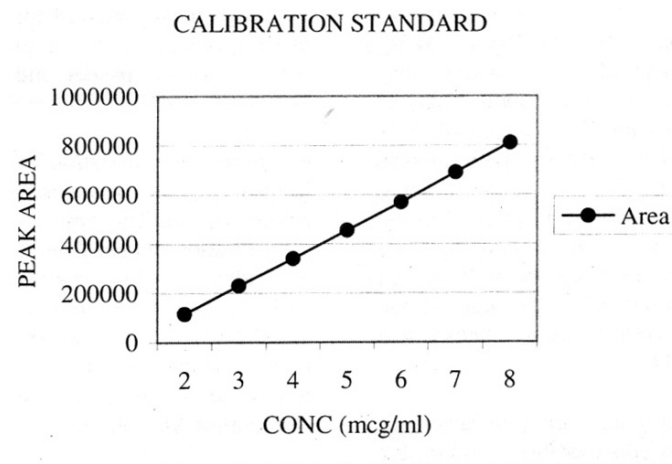


Figure 1

- D.** Flow Rate: 1 ml/min
E. Injection Volume: 20p.l injected with the help of SGE rheodyne syringe.
F. Wavelength Of Detection: UV 294nm
G. Sample Preparation: One ml Plasma was added to 1 ml Trichloroacetic acid and Vortexed using Whirl Mixer, and centrifuged for 10 minutes at 5000rpm.on Hereus centrifuge (Labofuge 200). Supernatant was injected in 20p,l Rheodyne Loop.
1. Retention Time: 13min

Calibration standard:

Different concentration of standard drug were made in Plasma. A sample from each concentration was treated in the same manner as test sample, using the same equipment and technical specifications. Readings were recorded and a standard curve was plotted. (Fig. 1)

RESULT AND DISCUSSIONS

The details about the volunteers and their AUC are indicated in Table 1.

Area under Plasma Concentration

Time Curve:

An estimation of area is required to

determine bioavailability and other Pharmacokinetics parameter. The most common method of estimating is the trapezoidal rule. The area of the trapezoid is equal to one half the product of the sum of the heights times the width. The area under a drug concentration in plasma versus time curve is approximated by the following equation. (Table 2, Figure 2)

$$\text{Area} = (1/2) (C_1 + C_2)(t_2 - t_1) + (1/2) (C_2 + C_3)(t_3 - t_2) + \dots + (1/2) (C_{n-1} + C_n)(t_n - t_{n-1})$$

Where C denotes drug concentration, t denotes time (Gibaldi M. 1991).

Applying this equation the Area under curve for Reference and test Product is

$$\text{AUC (Reference)} = 16.79$$

$$\text{AUC (Test)} = 16.987$$

BIOAVAILABILITY

The relative bioavailability of two products given at the same dosage level and by the same route of administration can be obtained with the following equation. (Macheras P. *et al.*, 1991)

$$\text{Relative Bioavailability} = (\text{AUC})_A / (\text{AUC})_B$$

TABLE 1

S. No.	Vol. Code	Age	Wt Kgs	Height (Cm)	B.P	AUC Test $\mu\text{g. hr/ml}$	AUC Ref $\mu\text{g hr/ml}$	C _{max} $\mu\text{g/ml}$	T _{max} , (hr)
1	V1	30	74	177.8	110/70				
-)	V7	35	75	182.8	120/80				
3	V3	28	67	167.6	110/80			<i>Test</i> 2.017 $\pm$ 0.4	<i>Test</i> 1.0
4	V4	27	65	170.0	110/80	16.987 $\pm$ 3.376	16.79 $\pm$ 1.794	<i>Ref</i> 1.85 $\pm$ 0.38	<i>Ref</i> 1.0
5	V5	32	57	176.0	120/80				
6	V6	38	63	175.2	100/70				
7	V7	34	72	177.8	110/70				

TABLE 2
Plasma drug concentrations at different time intervals for both test
and Reference Formulations

Time (hrs)	LEFLOX (TEST)									
	Concentration (µg/ml)							MEAN	STDEV	Conf. Interval
	S1	S2	S3	S4	S5	S6	S7			
0.5	1.62	1.2	1.81	1.81	1.49	0.49	0.29	1.24	0.62225	1.24±1.24
1	2.45	2.02	2.09	1.95	1.9	1.87	1.84	2.017	0.20974	1.88±.418
1.5	1.69	1.03	0.9	0.92	1.39	1.17	1.18	1.18	0.28034	1.223±.56
2	1.23	0.9	0.85	0.86	1.06	0.91	1.13	0.991	0.14871	.991±.28
3	1.071	0.88	0.84	0.8	0.9	0.79	0.85	0.875	0.09466	.875±.188
4	0.71	0.77	0.64	0.7	0.73	0.62	0.73	0.7	0.05292	.7±.104
8	0.677	0.65	0.63	0.6	0.72	0.59	0.54	0.629	0.06005	.629±.12
12	0.6	0.54	0.46	0.54	0.59	0.5	0.47	0.52	0.05475	.52±.108
24	0.545	0.41	0.34	0.43	0.43	0.45	0.4	0.42	0.06194	.42±.12
30	0.498	0.23	0.26	0.2	0.24	0.25	0.28	0.27	0.09943	.27±.198

Time hrs	CRAVIT (REFERENCE)									
	Concentration µg/ml							Mean	STDEV	Conf. Interval
	S 1	S2	S3	S4	S5	S6	S7			
0.5	1.02	1.01	0.84	0.87	0.87	0.78	0.75	0.8771	0.1042	.877±0.2
1	1.55	2.08	1.87	1.61	1.89	1.93	2.02	1.85	0.1991	1.85±.38
1.5	1.01	1.3	0.95	1.14	1.16	1.02	1.21	1.1129	0.1246	1.112±.24
2	0.86	0.93	0.9	0.9	0.94	0.88	0.9	0.9014	0.0273	.901±.04
3	0.75	0.85	0.79	0.79	0.84	0.8	0.8	0.8024	0.0337	.80±.066
4	0.63	0.66	0.68	0.69	0.7	0.72	0.72	0.6854	0.0326	.68±.064
8	0.56	0.55	0.58	0.66	0.64	0.6	0.66	0.6071	0.0464	.607±.092
12	0.5	0.49	0.5	0.57	0.56	0.51	0.59	0.5314	0.0406	.53±.08
24	0.45	0.43	0.44	0.46	0.51	0.41	0.52	0.46	0.0408	.46±.08
30	0.38	0.32	0.28	0.27	0.45	0.39	0.45	0.3629	0.0748	.36±.14

So the Relative bioavailability of Test vs
Reference Product of Levofloxacin is

$$16.987/16.79 = 1.01$$

or 101.1%.

STATISTICAL ANALYSIS

Analysis of Variance (ANOVA):

An analysis of variance is a statistical
procedure used to test the data for difference

within and between treatment and control group. A bioequivalent product should produce no significant difference in all Pharmacokinetics parameter tested (Shargel L

and Andrew B.C., 1999). Following are the layout for testing the ANOVA for these products. (Table 3)

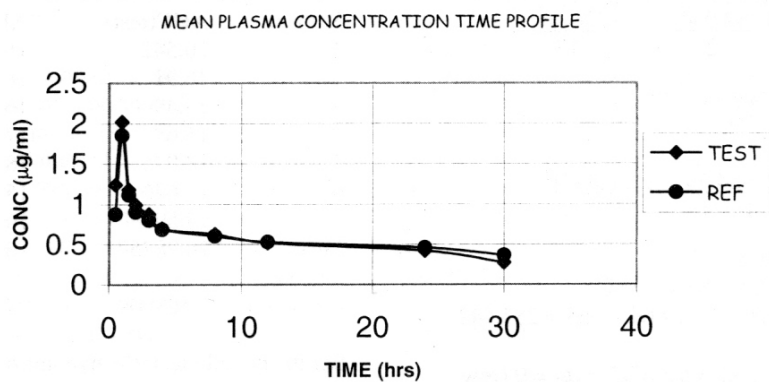


Figure 2

TABLE 3

Source of Variation	Sum of Square	Degree of Freedom	Mean Square	Computed F
Row Means	SSR	$r - 1$	$S^2_1 = SSR / r - 1$	$f_1 = S^2_1 / s^2_3$
Column Means	SSC	$c - 1$	$S^2_2 = SSC / c - 1$	$f_2 = S^2_2 / S^2_3$
Error	SSE	$(r - 1)(c - 1)$	$S^2_3 = SSE / (r - 1)(c - 1)$	
Total				

Subject	Formulations		Total
	A (Test)	B (Reference)	
	AUC	AUC	
1	20.542	16.182	36.724
2	16.91	16.642	33.552
3	15.49	16.57	32.06
4	16.68	17.633	34.313
5	18.075	18.484	36.559
6	15.626	15.559	31.185
7	15.59	16.464	32.054
Total	118.913	117.534	236.447

So applying the sum of squares computational formula;

$$SST = (20.542)^2 + (16.91)^2 + \dots + (16.464)^2 = 4293.66$$

$$SSR = \frac{(36.724)^2 + (33.552)^2 + \dots + (32.054)^2}{2} - \frac{(236.447)^2}{14}$$

$$= 4008.065 - 3993.37$$

$$= 14.69$$

$$SSC = \frac{27954.54}{1} - \frac{(236.447)^2}{14}$$

$$= .058$$

$$SSE = SST - SSR - SSC$$

$$= 300.29 - 14.69 - .058 = 285.542$$

1. $H_0(1) : \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$ (row effects are zero)
2. $H_0(2) : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$ (column effects are zero)
3. $H_1(1)$: At least one of the α is not equal to zero
4. $H_2(2)$: At least one of the β is not equal to zero
5. Critical Region: (a) $f_1 > 4.28$ (b) $f_2 > 5.99$
6. Result:
 - a. Accept H_0 that there is no difference in the average AUC when different subjects were used.
 - b. Accept H_0 that there is no difference in the average AUC when two different formulations were used.

Note: SST = Total Sum of Square
 SSC = Sum of Square of Column
 SSR = Sum of Square of Rows
 SSE = Sum of Square of Errors

Paired T Test:

One of the other method to evaluate the Bioequivalence of drug product is to use a simple t test to evaluate the Plasma concentration time curve (i.e Cmax, Tmax, AUC) (James E. De M. 1999).

Subject	AUC(test)	AUC(Ref)
1	20.542	16.182
2	16.91	16.642
3	15.49	16.57
4	16.68	17.633
5	18.075	18.484
6	15.626	15.559
7	15.59	16.464
Mean	16.98757	16.7906
Standard Deviation	1.688909	0.89777

For the paired t test, the hypothesis would be:

$$H_0: \mu = 0$$

$$H_1: \mu \neq 0$$

The decision rule, based on 0.05 Level of Significance is to reject H_0 if $t > t_{12}(.025) = +2.179$ or $t < -t_{12}(.025) = -2.179$.

The statistical analysis would be as follows

$$S^2_p = \frac{(n_1 - 1)S^2_1 + (n_2 - 1)S^2_2}{n_1 + n_2 - 2}$$

$$S = \text{Standard Deviation}$$

$$n = \text{No of Subject}$$

$$S^2_p = \frac{6(1.68)^2 + 6(.89)^2}{12}$$

$$S^2_p = 1.80$$

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S^2_p}{n_1} + \frac{S^2_p}{n_2}}}$$

$$T = 0.26$$

Source of Variation	Sum of Square	Degree of Freedom	Mean Square	Computed F
Row Means	14.69	6	2.44	$f_1 = .051$
Column Means	.058	1	.058	$f_2 = .012$
Error	285.54	6	47.59	
Total	300.288	13		

Where X is mean and T is for T distribution

The result is that we accept null hypothesis because the t value is less than our critical value of -2.179. Therefore it is concluded with 95% confidence that there was no difference between Bioavailability of two formulations

All fluoroquinolones are well absorbed; reach peak serum concentrations within 1-2 hr (Sparfloxacin in 4-5 hr), and have high distribution volumes (1.5-4.5 liters/kg) typical of the class. Protein binding is generally <50%, although somewhat higher (70%) for Trovafloxacin. After oral administration the fluoroquinolones produce levels well in excess of those required for clinical efficacy in genitourinary, gynecological, skeletal, and other important tissues and fluids (James E. De. M 1999).

A study by Davis & Bryson in 1994 showed that Levofloxacin penetrates rapidly into body tissues. Following oral administration of 50-200 mg Levofloxacin, mean maximum plasma concentrations ranging from 0.57 to 2.04 mg/l were achieved within 0.8 to 2.4 hours; these were linearly related to dose (Davis R and Bryson H.M., 1994). We observed that Levofloxacin 250mg have average Cmax of 2.017 ± 0.4 in test formulation and 1.38 ± 0.38 in reference formulation. Tmax of both reference and test formulation is 1 hr.

In the present study a single 250 mg dose of Leflox and Cravit were found to have similar bioavailabilities based on comparative values of C-max, AUC, and T-max. Mean C-max and AUC values $\pm$ standard deviation following a single 250 mg dose of Leflox is 2.017 ± 0.4 mcg/ml and 16.987 ± 3.376 mcg.h/ml, respectively, compared to the same dose of cravit, given to the same volunteers after a one week wash out period, their Mean C-max and AUC values $\pm$ standard deviation are found to be 1.85 ± 0.38 mcg/ml and 16.79 ± 1.794 mcg.h/ml, respectively. The T-max values for both test and reference are found to be 1.0 hour.

Levofloxacin is rapidly absorbed after single and multiple once-daily doses, with an apparently large volume of distribution. In a previous study it was found that the mean C-max and AUCO-24 values $\pm$ SD following a single 750 mg dose were 7.13 ± 1.44 mcg/ml and 71.3 ± 10.3 mcg.h/ml, respectively, compared to 8.60 ± 1.86 mcg/ml and 90.7 ± 17.6 mcg.h/ml at steady state. Following the single 1 gm dose, mean C-max and AUCO-24 values $\pm$ SD were 8.85 ± 1.86 mcg/ml and 95.4 ± 16.0 mcg.h/ml, respectively; corresponding values at steady state were 11.8 ± 2.52 mcg/ml and 118 ± 18.9 mcg.h/ml. These indicate modest and similar degrees of accumulation of Levofloxacin for the two dose levels upon multiple dosing (Gerding and Hitt., 1989).

One of the previous studies also indicated that the mean values of C-max and AUC, for a 100 mg dose after meal were found to be 1.22 ± 0.08 mcg/ml and 7.46 ± 0.34 mcg.h/ml respectively, and same dose under fasting were found to be 1.36 ± 0.16 mcg/ml and 10.42 ± 0.43 mcg.h/ml respectively, and for a dose of 200 mg after meal were found to be 2.04 ± 0.21 mcg/ml and 19.88 ± 1.15 mcg.h/ml respectively. T-max for the dose of 100 mg after meal, 100 mg under fasting and 200 mg after meal were found to be 0.92 ± 0.31 h, 0.82 ± 0.18 h and 1.48 ± 0.31 h, respectively (Nakashima, M 1992). The maximum serum concentration and area under the serum concentration curve were dose dependent.

The factors which can reduce the bioavailability of Levofloxacin include Interactions with Dietary Supplements and Interactions with Foods and Other Compounds. Taking magnesium supplements at the same time as Levofloxacin can reduce the intestinal absorption and thus the effectiveness of the drug. Consequently, nutritional supplements or antacids containing magnesium, if used, should be taken two hours before or after taking Levofloxacin. Taking iron supplements concomitantly with Levofloxacin can reduce the absorption and thus the effectiveness of the drug (Fish D.N.

and Chow A.T., 1997). Therefore, nutritional supplements containing iron, if used, should be taken two hours before or after taking Levofloxacin. Caffeine may have an intensified effect in people taking Levofloxacin. Drugs similar to Levofloxacin have been shown to cause caffeine to persist longer in the blood (Rodvold K.A. and Neuhauser M. 2001). However, the effects of Levofloxacin on caffeine blood levels or symptoms of caffeine ingestion have not been studied. To eliminate all these factors the Levofloxacin was given in fasting state in the present study.

Rodovold and Hauser showed that the Levofloxacin had excellent oral bioavailability (90-99%), extensive penetration into tissues and body fluids, and an elimination half-life 6-12 hrs.

Recent pharmacokinetic data showed that oral absorption was highest for Levofloxacin (99-100%), and 500 mg oral Levofloxacin achieved a much higher initial concentration than either Sparfloxacin (400 mg) or Ciprofloxacin (500 mg, b.i.d.), with a slow drop in concentration over 24 h. The C(max) achieved after an oral 250-mg dose is 2.48 for Levofloxacin ($p < 0.01$) having AUC 17.9 (Lode H. 2001).

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